BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 022 · Client–Server and Peer-to-Peer Networks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What distinguishes a LAN?

Q2. How is an item accessed in a 1D array?
A. Using its font
B. Using an index
C. Using a file extension
D. Using a network address

Q3. What is the difference between input and output?

Q4. A ticket program checks ages. “A few successful tests prove it has no faults.” Correct this. How should it be tested?

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I Do · Read new knowledge

Key vocabulary

Client — Device that requests a network service.

Server — Device that provides a service or resource.

Client–server — Network with clients and separate service-providing servers.

Peer-to-peer — Devices directly sharing resources as peers.

Central backup — Data copies managed from a central location.

Permissions — Rules controlling which users can access resources.

New learning

Client–server: Clients ask for services; servers provide them. The two computers have different roles.

Peer-to-peer: Peers can each share and request files without needing a dedicated central server.

Client–server benefits: Central permissions, user management and backups can help larger organisations; dedicated servers need investment.

Peer-to-peer trade-offs: A small group can share directly at low cost, but there is no central backup and files may be unavailable if the sharing device is switched off.

Worked example

A secondary school needs user logins, shared files and backups. Explain TWO advantages of client–server networking. [4 marks]

1. The school needs controlled accounts and reliable shared files.

2. A central server can manage logins/permissions [1], making access control easier [1].

3. Files are backed up centrally [1], helping restore data after failure [1].

4. Two separate advantages, each supported by a consequence, gain four marks.

We Do · Practise with guidance

Guided 1

A small art group with four laptops shares files directly. Explain why peer-to-peer could be suitable. [4 marks]

Worked method: It needs no dedicated server [1], keeping setup cheap [1]. Laptops share files directly [1], which is simple for a small group [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Guided 2

A health clinic stores confidential files. Explain two advantages of a managed client–server network. [4 marks]

Worked method: Central permissions restrict access [1], supporting confidentiality [1]. Central backups [1] let records be restored after a failure [1].

Guided 3

A peer shares a file but switches off their computer. Explain one disadvantage of peer-to-peer and another related to security. [4 marks]

Worked method: The resource becomes unavailable [1], interrupting other users [1]. Without central security management [1], controlling access consistently becomes harder [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the message.

```
permission = true
if permission then
    print("Access granted")
else
    print("Access denied")
endif
```

predict

Question 1

State the role of a server. [1 mark] Source: Adapted from your Paper 1 revision booklet, section 1.3.1.

Question 2

A community drama group directly shares sound files between its rehearsal laptop and another computer, without a dedicated server. Explain one benefit and one drawback of peer-to-peer networking. [4 marks] Source: Original J277-style practice, not an OCR past-paper original.

Question 3

A leisure centre has over 100 staff accounts and shared booking records. Compare client–server and peer-to-peer networking for this organisation. [6 marks] Source: Original J277-style practice, not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

All peer-to-peer devices must request files from one dedicated central server.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It connects devices across a small area, such as one school site.

Q2. B — Using an index

Q3. Input enters a program; output is information the program produces.

Q4. Test valid and invalid inputs. Compare actual and expected results to find faults. A few successful tests cannot prove there are no faults.

I Do / We Do pages · Guided practice

Q1. It needs no dedicated server [1], keeping setup cheap [1]. Laptops share files directly [1], which is simple for a small group [1].

Q2. Central permissions restrict access [1], supporting confidentiality [1]. Central backups [1] let records be restored after a failure [1].

Q3. The resource becomes unavailable [1], interrupting other users [1]. Without central security management [1], controlling access consistently becomes harder [1].

You Do pages · Code First

```
permission = true
if permission then
    print("Access granted")
else
    print("Access denied")
endif
```

Expected result: Access granted

You Do pages · Independent questions

Q1. Provides services/resources to clients [1].

Q2. No dedicated server needed [1], reducing cost [1]. The other laptop can be off [1], preventing shared-file access [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Client–server: central server provides services [1], enabling permissions [1] and central backups [1]. Peer-to-peer: devices share directly [1] with cheap setup [1], but central control and backups are harder or files may be unavailable when peers are off [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: